



Pablo Martínez Agulló

PhD in Physics

Data Science - Artificial Intelligence

AI specialist with a strong foundation in Machine Learning across both research and applied settings.

I have trained and deployed deep learning models, and developed production-ready applications leveraging Generative AI through foundation models. My work spans from scientific research at CERN to delivering business-driven AI solutions at Mercadona IT, combining technical rigor with innovation to create real-world impact.

EDUCATION

2017-2024

Doctorate in Physics | University of Valencia (Spain)

2016-2017

MSc in Advanced Physics | University of Valencia (Spain)

2015-2016

Erasmus | RTWH Aachen University (Germany)

2011-2015

Degree in Physics | University of Valencia (Spain)

EXPERIENCE

October 2024 - Present



Mercadona | Artificial Intelligence Engineer (Spain)

- AI developer in the Innovation team (IT department)
- Designed and deployed AI-powered solutions in a production environment, leveraging both Generative AI (Gemini, Vertex AI) and classic ML techniques.
- Built a RAG-based chatbot to navigate internal policy documents using LangChain, FAISS, DialogFlow CX, and foundation models.

December 2017 – May 2024



IFIC/CERN | Researcher (Spain/Switzerland)

- Analysis of large volumes of data in ATLAS collaboration
- Use of advanced statistical techniques for data interpretation
- Development, optimisation, and integration of supervised Deep-Learning models
- Translation of scientific theories into algorithms
- Collaborative software development
- Use of version control systems
- Presentation of results at national and international conferences
- Data-taking responsible for the ATLAS detector
- Development of front-end and back-end web service to monitor the performance of the ATLAS detector
- Development and integration of tools for detector simulations

February 2020 – September 2021



University of Valencia | Lecturer (Spain)

- Teaching and assessment of the courses Electromagnetism and Physics Laboratories

June 2016 – September 2016



ESA | Internship (Netherlands)

- Internship at the European Space Agency (ESA) and Leiden University
- Study of the impact of outreach multimedia using an NLP-based application

RESEARCH STAYS

May – July 2022

CERN (Geneva, Switzerland)



September – December 2018

CERN (Geneva, Switzerland)



March – May 2018

CERN (Geneva, Switzerland)



OUTREACH

- Speaker and host of the event *70 años del CERN* (800+ attendees, 2025)
- Contributor to the physics podcast *Oscilador Armónico* (2023 - 2024)
- Speaker/host of the public event *10º aniversario del descubrimiento del bosón de Higgs* (700+ attendees, 2022)
- High school talks: *¿De qué está compuesta la realidad?* (2022-2023)
- Video *LHCRUN3* (21k+ views, 2022)
- Organizer of the science fairs *Expociencia* and *Experimenta* (2019-2022)

HARD SKILLS

Programming & Languages:

Python (expert), Bash, C++, SQL, HTML/CSS,

AI & Machine Learning:

Supervised ML and Deep Learning (XGBoost, Scikit-learn, PyTorch, Keras, TensorFlow), NLP, Foundation Models, Generative AI, Prompt Engineering, OCR with LLMs, RAG pipelines

Data Science & Analytics:

Pandas, NumPy, SciPy, Matplotlib, SQL, ROOT, TMVA, BeautifulSoup, Statistical Analysis, Feature Engineering, Data Visualisation & Communication

Frameworks & Tools:

FastAPI, Flask, Django, LangChain, FAISS, DialogFlow CX, Pyomo, Poetry, Git/GitHub/GitLab, Docker, Kubernetes, CI/CD (Spinnaker, CloudBees), Google Cloud Platform (Vertex AI, Cloud Functions, Cloud Storage)

Scientific & Applied Experience:

Conversion of scientific theories into algorithms, advanced statistical modelling, large-scale simulations, domain-specific AI solutions for research and business, and full-cycle project delivery from PoC to production

CONTACT



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GitLab: @martinpa



GitHub: @MartinezAgullo

SCOLARSHIPS

- 2019: Scholarship from the Valencian Government (GVA) for the hiring of research personnel
- 2017: Severo Ochoa Collaboration Scholarships at CSIC Centers of Excellence at IFIC

SOFT SKILLS

- Collaboration & cross-functional teamwork
- Autonomy
- Leadership
- Very good communication and interpersonal skills
- Creative problem-solving
- Large experience in international environments
- Proactive & eager to learn
- Time and deadline management
- Critical & strategic thinking

LANGUAGES

- Spanish :: Native
- English :: Professional
- Catalan :: Native
- German :: Basic